

THARIQ T

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PROFESSIONAL SUMMARY

A problem-solving undergraduate in Electronics and Communication Engineering from TKM College of Engineering with a Minor in Machine Learning. Hands-on experience in embedded systems hardware and software integration, analog/digital circuits and real-time signal processing. Demonstrated ability to rapidly learn new systems, collaborate cross-functionally, and deliver results on time. Aspire to apply technical knowledge and analytical thinking to contribute meaningfully to organisational success while continuously growing both personally and professionally.

EDUCATION

Bachelor of Technology — Electronics and Communication Engineering

TKM College of Engineering, Kollam, Kerala | Minor in Machine Learning | CGPA: 8.58 / 10.0 | 2022 – 2026

TECHNICAL SKILLS

Programming & Scripting Languages: Python

Hardware & Protocols: I2C, SPI, UART

Analog & Digital Electronics: Op-Amps, data converters, signal amplifiers, modulation, digital logic design, CMOS fundamentals

Embedded Platforms: Arduino Uno, Raspberry Pi, ESP8266, Jetson Nano

Machine Learning / AI: TensorFlow, Keras, PyTorch, ONNX, BiLSTM, CNN, DistilBERT, RandomForest, YOLO

Lab Skills: Oscilloscope usage, soldering, circuit debugging, sensor integration

Tools & Platforms: GitHub, VS Code, OpenCV

INTERNSHIP EXPERIENCE

Hardware & Embedded Systems Intern

Nest Digital, Trivandrum | Feb 2026 – Mar 2026

- Designed and presented an industrial-level automated, customisable currency-sorting robotic arm integrating mechanical and electronic subsystems.
- Developed embedded C firmware for motor control and sensor feedback loops on a microcontroller platform.
- Collaborated with cross-functional stakeholders to iterate on design specifications and deliver on-time.

Embedded Systems & Computer Vision Intern

Centre for Development of Advanced Computing (C-DAC), Trivandrum | Jun 2024 – Jul 2024

- Built a facial-recognition attendance system on Raspberry Pi 3 using Python and OpenCV; integrated I2C/SPI peripherals.
- Developed an object-recognition system for visually impaired users, combining real-time inference with audio feedback.
- Gained hands-on experience debugging hardware–software interfaces and sensor data pipelines.

Signal Processing & Broadcasting Intern

All India Radio, Trivandrum | May 2023

- Studied RF broadcasting equipment covering signal transmission/reception, amplifier tuning, and AM/FM modulation circuits.

ACADEMIC PROJECTS

Speed Sign-Board Detection & Speed Control System

Technologies: YOLOv11n, Raspberry Pi, Python

- Built a real-time computer vision pipeline for traffic-sign detection and automatic speed regulation.
- Interfaced deep-learning inference outputs with embedded motor-control logic — bridging software and hardware layers.
- The dataset is prepared from scratch under varied environmental conditions

Agentic AI-Based Personalised ADAS Vehicle

Technologies: Raspberry Pi 5, Python, PWM, ESC, BLDC Motor, HC-SR04, DistilBERT, Vosk

- Designed and built a prototype ADAS car: integrated BLDC motor, ESC, and ultrasonic sensors via hardware PWM on Raspberry Pi 5.
- Implemented real-time I2C/SPI-based sensor fusion for obstacle detection and adaptive speed control.
- Developed embedded firmware in Python for PWM timing and ESC arming sequences.
- System maps driver emotion (voice) to driving modes; demonstrates ability to rapidly adopt new embedded platforms.

Humanoid Robot — AI Voice-Assisted Welcome Robot

Technologies: Raspberry Pi 5, Servo Motors, Face Recognition, Python

- Programmed 12+ servo motors via I2C PWM driver board; implemented face-recognition pipeline for personalised greetings and actions.
- Debugged hardware signal integrity issues using oscilloscope and logic analyser.

Smart Helmet Safety System

Technologies: ESP8266, Gyroscope, Alcohol Sensor, GSM Module

- Designed analog sensor circuit (gyroscope + alcohol sensor) feeding into an ESP8266 microcontroller for accident detection and SOS alerts.
- Soldered and assembled PCB components; validated circuit operation with bench instruments.

Personalised Sleep-Stage Classification using Deep Learning

Technologies: BiLSTM, TensorFlow, Python

- Engineered acoustic and physiological features from raw datasets; trained a BiLSTM achieving high-accuracy multi-class classification.
- Optimised model for low-latency inference — applicable to automotive drowsiness-detection use cases.

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL — Joy of Computing Using Python, IIT Madras
- C-DAC Internship Certification, Trivandrum
- All India Radio Internship Certification, Trivandrum
- Best Mini Project Award — TKM College of Engineering

LEADERSHIP & EXTRACURRICULAR

- Executive Committee Member — Drone and Robotics Application Centre (DARC), TKM College of Engineering
- NSS Volunteer — TKM College of Engineering (community service & event organisation)
- Peer Tutor — Mathematics and Physics (strong communication and interpersonal skills)

LANGUAGES

English | Malayalam